

EDITORIAL

SMS | Strategic Entrepreneurship Journal



WILEY

Innovating for impact: Harnessing business model innovation to tackle grand challenges

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Abstract

Research Summary: This Special Section of the *Strategic Entrepreneurship Journal* explores how business model innovation (BMI) can address grand challenges (GCs), including artificial intelligence (AI), business and global unrest, and climate change through a proposed “ABC” classification and “BMI-GC” concept. Highlighting BMI's role as a boundary-spanning, collaborative approach to innovation, we demonstrate its potential to drive sustainable solutions by rethinking value creation, delivery, and capture mechanisms. We examine how BMI may foster resilience and help firms and organizations adapt to global volatility, integrate ethical AI, and advance sustainability in business ecosystems. We offer a roadmap for future inquiry into BMI-GC, including measurement and assessment, digital transformation, and collaborative ecosystems. By embracing these research pathways, scholars can contribute to a deeper theoretical and practical understanding of BMI to address grand challenges.

Managerial Summary: We investigate how business model innovation (BMI) can help tackle pressing global challenges

(GCs), including artificial intelligence (AI), business and global unrest, and climate change, captured in a proposed “ABC” classification and “BMI-GC” concept. We illustrate how BMI can help firms and organizations create sustainable, resilient, and future-proof solutions. Practical strategies include integrating ethical AI, embedding flexibility in business models to withstand volatility, and adopting sustainable practices that align with regulatory and market demands supported by BMI. The paper identifies key opportunities for businesses to rethink how they create and deliver value, and foster collaboration across ecosystems. Through the ABC classification and BMI-GC concept, we show how BMI can help enhance competitive advantage while responding proactively to the complexities of a rapidly changing world.

KEYWORDS

AI, business model innovation, climate change, grand challenges, resilience

1 | INTRODUCTION

Business model innovations entail holistic changes to how business is conducted and often impact multiple organizational dimensions and stakeholders (Amit & Zott, 2021; Klein et al., 2025; Spieth et al., 2025; Tucci, 2022). Business models can be defined as boundary-spanning activity systems, centered on the focal firm but spanning focal firm boundaries to include activities performed by customers, suppliers, and partners that, taken together, allow the firm to create and capture value (Amit & Zott, 2001; Zott & Amit, 2010). Building on this conceptualization, we define new business models as systems of interconnected transactions and activities centered on a focal organization and designed to serve markets or solve problems in ways that are either new to the world, new to a market, or simply new to the focal organization (Massa & Tucci, 2013; Snihur & Bocken, 2022; Snihur & Zott, 2020; Spieth et al., 2025). Business model innovation strategies are among the key categories of strategic issues that require attention from top executives and entrepreneurs (Amit & Zott, 2021; Eklund & Mannor, 2021).

Our objective with this Special Section is to link our improved conceptual and empirical understanding of business models with the important notion of *grand challenges*, the urgency of which has become increasingly apparent. Grand challenges relate to those presented by the UN Sustainable Development Goals (SDGs), such as eradicating poverty, reducing inequality, increasing access to clean water and sanitation, and providing affordable clean energy. They also include the transition to a circular economy (Geissdoerfer et al., 2017; Klein et al., 2023). Grand challenges cannot be resolved by individual actors alone and require collaboration and coordination within and across activity systems (Ferraro et al., 2015; George et al., 2016). While typically addressed by national and supranational institutions and NGOs, for-profit organizations are increasingly entering the territory of resolving grand challenges through innovation, including business model innovation (Ritala, 2024).

To systematically examine these challenges, and what kind of innovative business model solutions could be associated with them, in this article we propose an “ABC” classification that encapsulates three pivotal categories of



grand challenges: artificial intelligence (A), business and global unrest (B), and climate change (C). Correspondingly, the top current global risks are (1) AI and misinformation (Neufeld, 2024); (2) a host of other factors that may be grouped together as business/global unrest, including societal and political polarization, the cost-of-living crisis, economic downturns, and disrupted supply chains for critical goods and resources (Allianz Commercial, 2024; World Economic Forum (WEF), 2024); and (3) climate-related risks (e.g., extreme weather patterns) (IPCC, 2018a, 2018b).

Focusing on these interconnected areas, our framework provides a structured approach to understanding and categorizing the most pressing global issues affecting businesses today. However, as entrepreneurship scholars, we know that challenges always present opportunities for aspiring entrepreneurs in two ways. First, strategic entrepreneurship initiatives, such as BMI at the base of the pyramid, or new social ventures with innovative business model designs, can be deployed to address (i.e., try to solve) grand challenges. Second, grand challenges could give rise to (i.e., foster and enable) business model innovations. Given the relevance and timeliness of the topic, and the lack of research at the intersection of business model design, innovation, strategic entrepreneurship, and grand challenges, we launched this Special Section.

In doing so, we build on the significant research on business models that has been performed over the past two decades (see Zott et al., 2011 and Massa et al., 2017, for reviews), and for which the *Strategic Entrepreneurship Journal* (SEJ) has been a prime outlet—see the *SEJ Special Issue on Business Models* 2015, and several *SEJ* articles on business models that have received *SEJ* Best Paper awards, such as Andries et al. (2013), Amit and Zott (2015), or Amit and Han (2017).

2 | LITERATURE BASE LINKING BMI TO GRAND CHALLENGES

To gain a thorough overview of prior research on the links between business model innovation and grand challenges, we conducted a comprehensive database search on EBSCOHost, ScienceDirect, and Web of Science (Bartels & Reinders, 2011), encompassing articles available through December 31, 2024. We conducted three separate database searches (details in the [Supporting Information](#) Appendix) on BMI research in the context of each of the three identified grand challenges: artificial intelligence (AI), business and global unrest, and climate change.

During the first step of the search process, we sourced 304 articles for BMI and AI, 27 for BMI and business/global unrest, and 60 for BMI and climate change. In the second step, duplicates were eliminated, and the eligibility of the remaining articles was assessed through a thorough analysis of the full texts. Eligibility criteria focused on whether the research centered on deploying business model innovations in the context of the three grand challenges, rather than merely mentioning the grand challenge as a tangential aspect. In step three, we augmented the remaining articles from the database searches—49 for AI, 17 for global unrest, and 12 for climate change—with additional articles identified through cross-referencing and manual searches. This added seven articles for AI, 26 for global unrest, and 11 for climate change, for a total of 122 articles. Our Supporting Information Appendices B.1–B.3 outline the characteristics of the final set of articles.

As shown in Figures 1–3, publication activities on BMI in the context of each grand challenge have significantly evolved over the years, with a steep incline in recent years. However, many of the recent academic conversations on the topics of interest for this Special Section have been conducted outside the leading mainstream entrepreneurship and strategy journals (e.g., Bocken et al., 2014). The purpose of this Special Section is to provide a forum for such discussions and encourage research at the intersection of BMI and global challenges using a strategic entrepreneurship lens.

In the following section, we summarize the articles in the Special Section, drawing selectively on them and from some of the articles identified in our literature review on the links between BMI and grand challenges to offer a framework for future research.

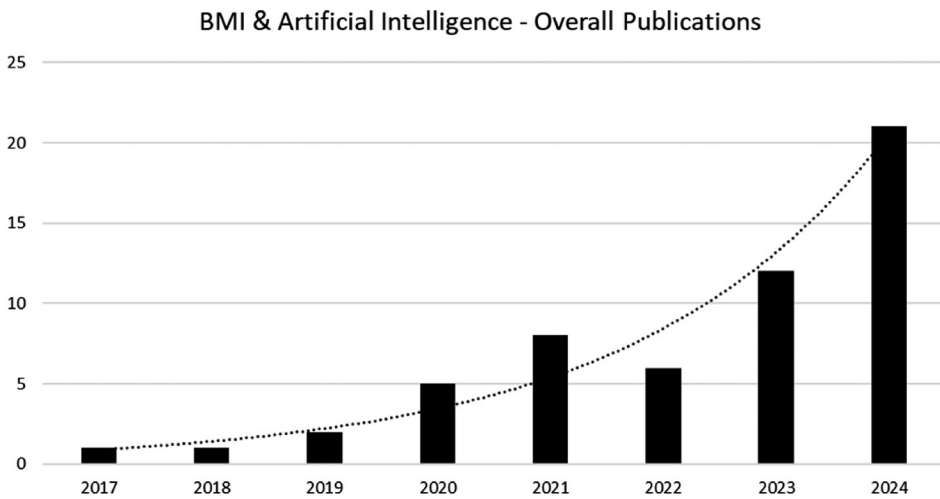


FIGURE 1 Literature on BMI & AI—overall publications.

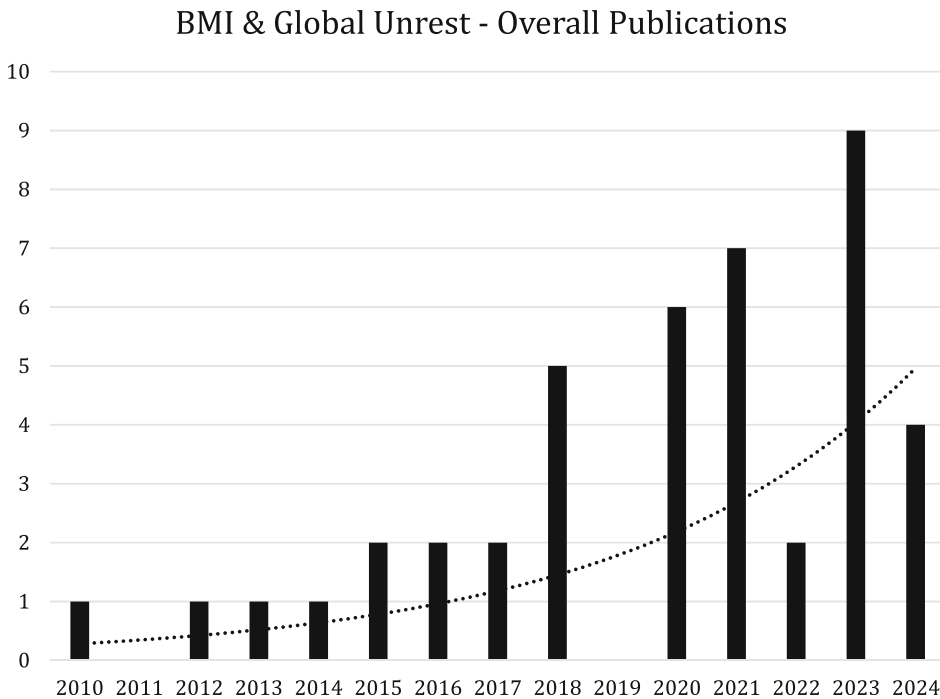


FIGURE 2 Literature on BMI and global unrest—overall publications.

3 | SPECIAL SECTION ARTICLES

This Special Section is designed to raise awareness about the importance of the topic, and about the associated high impact that related work is likely to have on both academia and practice. It was conceived in 2021 and launched in 2022 with a due date of April 2023 for the submission of articles. We received a total of 20 manuscripts, of which seven were sent out for review and 13 were desk rejected for reasons of quality or lack of fit with the Special



BMI & Climate Change- Overall Publications

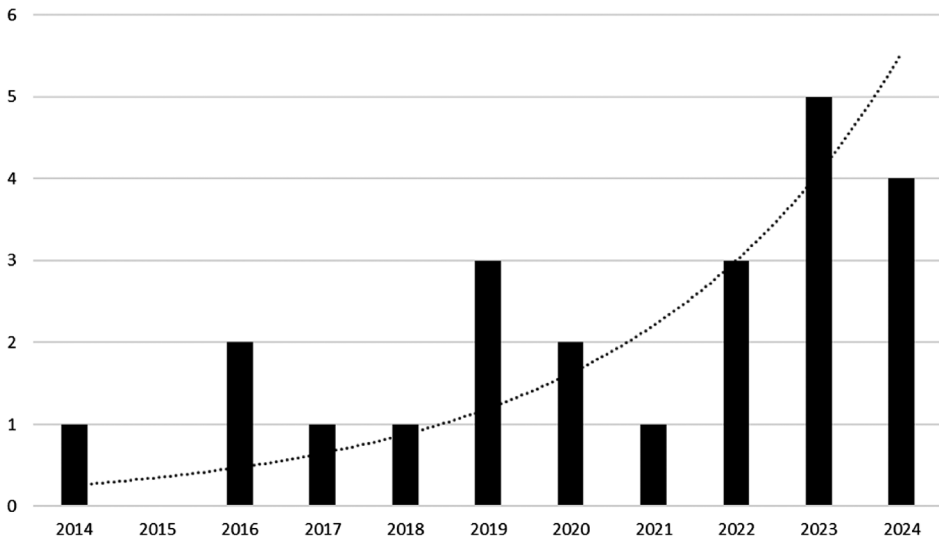


FIGURE 3 Literature on BMI and climate change—overall publications.

Section. After conducting a thorough multi-stage review process, we arrived at three final acceptances, which are summarized below.

The first article, “Stairway to impact or highway to failure? A cognitive perspective on business model design processes in nascent sustainable ventures,” by Esau et al. (2025), investigates how new ventures develop sustainable business models to address ecological and social challenges while maintaining profitability. Employing a longitudinal case study approach, the authors analyze six nascent ventures to identify two cognitive trajectories: *narrow* and *paradoxical*. Narrow-framed teams exhibit focused “either/or” thinking, often leading to unbalanced business models that struggle to scale or survive. In contrast, paradoxical teams embrace “both/and” thinking, integrating social, ecological, and commercial goals that foster innovative and adaptable business models. The study highlights the importance of collective cognition, demonstrating how cognitive frames within founding teams shape venture outcomes. Key contributions include a dual-path model for sustainable business model development and insights into the role of cognitive diversity and team dynamics in addressing tensions between competing goals. These findings offer a more nuanced understanding of the processes that determine success or failure in sustainable ventures that contribute to resolving the climate change challenge by tackling diverse issues such as reusable packaging to cut waste and cargo bike transport and sustainable products to reduce an offering's overall carbon footprint.

The second article selected for inclusion in this Special Section is “Navigating grand challenges: How environmental dynamism shapes robust action and business model Innovation” by Talebian et al. (2025). The authors develop a conceptual framework to explain how incumbent firms engage in BMI when addressing grand challenges under varying levels of environmental dynamism. Drawing from robust action theory, the authors identify three mutually reinforcing action strategies—participatory architecture, multivocal inscription, and distributed experimentation—that enable firms to generate and sustain collective responses to complex societal problems. The framework links these strategies to four types of BMI (evolutionary, adaptive, focused, and complex), showing how the degree of environmental change (regular, rapid, or disruptive) shapes firms' engagement in robust action and, ultimately, the novelty and scope of their BMI. Using theoretical arguments and illustrative case examples, the paper sheds light on the tensions that firms face in balancing exploration and coordination as they respond to social and environmental imperatives. Key contributions include advancing theory on grand challenge BMI in incumbent firms,

integrating insights from robust action into strategic entrepreneurship, and offering a nuanced understanding of how contextual factors influence organizational responses to GCs.

The final paper, “Digital technology-based business model design and innovation to address grand challenges: A process model,” by Wang et al. (2025), examines how a social enterprise in China utilized remote sensing data to tackle diverse grand challenges. The authors present a four-stage model—accessing, embedding, strengthening, and expanding—that enables ventures to design and innovate business models capable of increasing both the scale and scope of their impact. The study highlights the venture's success in transitioning from enhancing agricultural insurance ecosystems to addressing challenges in forestry, ecology, and renewable energy. Key contributions include elucidating the process through which digital technology enables BMI across multiple ecosystems and identifying the critical role of technological affordances in broadening a firm's impact. This work advances the literature on ecosystem strategies and digital business models, offering practical insights for firms aiming to achieve scalable and diverse social and environmental objectives.

Taken together, the three papers in this Special Section showcase the richness of interesting issues at the intersection of strategic entrepreneurship, BMI and grand challenges, and underline the potential for conducting meaningful and impactful research.

4 | ABC CLASSIFICATION AND BMI-GC CONCEPT

ABC refers to three pivotal categories of grand challenges: artificial intelligence, business and global unrest, and climate change. To begin with category A: Misinformation and the possible adverse outcomes of AI are high in the ranking of the severest risks facing humankind today (WEF, 2024). AI may eventually even threaten human existence and do so at speed if and when it happens. ChatGPT experienced flash growth of millions of users within a period of weeks—exhibiting almost vertical technology adoption curves that had been unseen in even the most recent history of technology development (Shrier, 2023). Such developments are hard to follow and enact by policy makers. This begs the question of how governments, businesses, and entrepreneurs should respond to a world in which the pace of change is several orders of magnitude faster than the typical pace of government intervention (Shrier, 2023; Tucci, 2018).

Category B refers to risks for businesses arising not only from global unrest and disruption due to polarization, economic uncertainty, and inflation, but also from disrupted supply chains due to wars, etc. “*The aftermath of the COVID-19 pandemic and ongoing Russia-Ukraine war has exposed cracks in societies that are being further strained by episodic upheaval. Yet the global system has thus far proved surprisingly resilient. A widely anticipated recession failed to materialize last year, and financial turbulence was quickly subdued, but the outlook remains uncertain*” (WEF, 2024, p. 12). Disrupted supply chains for food and energy remain important concerns due to wars and unrest, magnified by climate pressures.

As for the C category, many businesses have already been directly impacted by the effects of climate change and disasters due to extreme weather. According to the WEF (2024), the top four global risks ranked are all climate related: extreme weather events, critical change to earth systems, biodiversity loss and ecosystem collapse, and shortages of natural resources. Such risks are already affecting current business practices and pose a severe risk to ongoing business operations globally. They are both physical (e.g., heatwaves, drought, rising sea levels, increased rainfall, reduced soil productivity) and transitional (e.g., regulatory requirements, taxation, and the behavior of investors and consumers) (European Central Bank (ECB), 2022).

In an era of rapid technological advancements and significant societal shifts, businesses face complex challenges that demand innovative responses (Voegtlin et al., 2022). The “ABC classification” developed here refers to global problems that need to be addressed through coordinated and collaborative efforts (George et al., 2016), and equally importantly through the actions of entrepreneurial actors.



TABLE 1 Exemplary research questions for BMI and AI.

	How can AI be leveraged for BMI?	How can BMI help counter AI threats & mitigate AI risks?	How can AI spur innovation in business models that help alleviate grand challenges?
BMI antecedents	What are the organizational antecedents necessary for leveraging AI to create new business models? (see Talebian et al., 2025)	How can BMI incorporate frameworks for ethical AI deployment to ensure transparency and reduce biases?	Given rapid changes in AI technology development, regulation, and ethical standards, how can firms design adaptive AI-driven business models that help alleviate any of the ABC grand challenges?
BMI outcomes	How can AI help startups develop innovative business models that scale rapidly?	What governance structures are required for managing the risks of misinformation, privacy breaches, and ethical dilemmas in AI-driven business models? (see Esau et al., 2025)	How can AI be used to develop new value propositions that address societal needs, such as health, education, or environmental monitoring? Or value propositions that balance profitability and social impact?
BMI processes	How can AI help with more effective and efficient ideation, iteration, and implementation for the development of innovative business models?	What are the challenges and trade-offs firms face when embedding AI ethics into BMI processes? (see Wang et al., 2025)	What AI-driven processes should entrepreneurs use (or avoid) to design business models that reduce poverty and inequality? How do digital technologies enable the development of impactful business models that address grand challenges? (see Wang et al., 2025)

We propose business model innovation for grand challenges or “BMI-GC” as a novel concept to guide future research and business action in this area. Business leaders and entrepreneurs are often guided in their BMI efforts by deeply intrinsic motivations such as the desire to preserve a natural environment or alleviate poverty (Sergeeva & Zott, 2025). BMI can be important to resolve grand challenges, as it involves fundamentally rethinking the system of activities that defines how a company creates, delivers, and captures value, leading to transformative impacts that extend beyond the organization to society and the environment (Amit & Zott, 2021; Klein et al., 2021; Snihur & Bocken, 2022). Connecting the ABC classification to BMI underscores the necessity for integrated solutions and offers a lens through which businesses can effectively navigate and mitigate grand challenges.

In a nutshell, BMI can be used to address grand challenges if designed the right way. At the same time, grand challenges can also pave the way for and enable BMI. We explain both pathways and explore these themes in greater depth in the next section, which is summarized in Tables 1–3 and represents our analytical framework. We will examine existing research (including the articles published in this Special Section) on how BMI can respond to and leverage these grand challenges effectively, identify gaps in the literature, and propose directions for future inquiry. We aim to shed light on the potential for businesses not only to adapt to challenges but also to drive meaningful change through innovative business models.

4.1 | Business model innovation and artificial intelligence

AI has been described as the study of methods for making computers behave intelligently— acting in ways most likely to achieve a goal or maximizing expected utility.¹ It includes technologies that allow machines to learn, reason, plan,

TABLE 2 Exemplary research questions for BMI and business and global unrest.

	How can instability be leveraged for BMI?	How can BMI help counter threats & mitigate risks due to instability?	How can instability spur innovation in business models that help alleviate grand challenges?
BMI antecedents	What contextual factors trigger BMI during periods of business and global unrest? (see Talebian et al., 2025)	What internal and external resources facilitate risk mitigation through BMI? (see Wang et al., 2025)	How do firms identify and mobilize resources to create new value propositions that address societal challenges during crises? (see Esau et al., 2025)
BMI outcomes	What competitive advantages do firms gain by leveraging business and global unrest for BMI?	What governance and structure adaptations are most effective in minimizing disruptions?	How do crisis-driven business models enhance sustainability and societal value? (see Wang et al., 2025)
BMI processes	How do firms conceptualize and execute BMI processes in response to business and global unrest? (See Talebian et al., 2025)	How can businesses seamlessly integrate crisis-responsive BMI to improve adaptability and resilience? (see Esau et al., 2025)	What iterative processes help businesses pivot to socially driven business models during crises?

TABLE 3 Exemplary research questions for BMI and climate change.

	How can climate change be leveraged for BMI?	How can BMI help counter climate-threats and -risks?	How can climate change spur innovation in business models that help alleviate grand challenges?
BMI antecedents	What climate risks are the major drivers for BMI? How can various organizational functions start addressing climate risks and opportunities more profoundly?	How can BMI drive virtuous cycles toward climate solutions, and break-down vicious ones? (see Wang et al., 2025) How does environmental dynamism shape BMI choices that address global challenges? (see Talebian et al., 2025)	What new BMI territories need to be explored triggered by climate change? How can climate risk can become the driver for BMI? (see Talebian et al., 2025)
BMI outcomes	How can companies create a competitive advantage by leveraging the need to mitigate and adapt to climate change? (see Esau et al., 2025)	How can BMI mitigate climate threats and risks? How can BMI lead to more climate risk resilient models? (see Wang et al., 2025) How can the real climatic impact of BMI be better anticipated and measured in business operations and value chains?	How can BMI help mitigate Climate Change in different sectoral contexts? (see Esau et al., 2025) How can BMI help solve multiple intertwined and wicked climate change issues?
BMI processes	How do companies incorporate climate adaptation and mitigation in their competitive BMI processes? How do companies develop processes for collaborative BMI to tackle climate change at a systemic level?	What BMI processes may support design against climate risks and threats? How do cognitive trajectories during BMI influence the development of sustainable business models? (see Esau et al., 2025)	What processes support design for climate adaptive and mitigating BMI?



perceive, and self-correct, thereby mimicking human intelligence. As such, AI has become a strategic design driver for business model innovation (BMI), enabling managers and entrepreneurs to create or redesign value-creating business models to strengthen competitive advantage (Amit & Zott, 2021).

For instance, a yacht engine manufacturer that previously sold diesel engines and outsourced maintenance added AI-based activities: sensing, data analysis, and predictive maintenance. This created new revenue streams through predictive maintenance services and subscription fees, enabling the servitization of a product-based business (Zott et al., 2024). Similar transformations are occurring across industries.

While AI has long been viewed as a general-purpose technology alongside blockchain or cloud computing, the release of large language models (LLMs) such as ChatGPT in late 2022 marked a turning point. Trained on vast data sets, LLMs predict the most likely next word rather than retrieve facts, and sometimes “hallucinate.” Despite lacking true knowledge, they can act as creative co-workers and tutors, exhibiting a form of co-intelligence (Mollick, 2024). LLMs may eventually advance toward artificial general intelligence (AGI) or even superintelligence—raising both opportunities and risks and fueling doubts about an AI hype (Bender & Hanna, 2025).

A review of the AI-BMI literature (see Supporting Information Appendices A1 and B1) revealed no studies in leading management journals addressing three key areas: (a) how AI enables business model innovation for competitive advantage; (b) how BMI can mitigate risks from AI use (e.g., misinformation, inequality); and (c) how AI-driven models can help tackle grand challenges. These important topics remain vastly underexplored, likely due to the novelty of the technology and the inherent time lag in scholarly research. With the ongoing rapid development and adoption of LLM-based tools, significant new research is expected.

Future studies could examine antecedents, processes, and outcomes of BMI. Regarding antecedents, Amit and Zott (2015, 2021) identify six contextual categories guiding BMI in practice: a focal firm's resources and capabilities, external trends and technologies, partners, business model templates, managerial and organizational goals, and stakeholder needs. For example, Sena and Nocker (2021) highlight how AI as an antecedent shapes market structures, technical standards, and ethical frameworks—encouraging experimentation with value propositions, partnerships, and AI integration. Although not focused solely on AI, Talebian et al. (2025) provide a contextual theory explaining why and how firms adopt different types of BMI under dynamic and uncertain environments—conditions strongly shaped by AI-driven technological change.

For outcomes, research could analyze specific choices of how AI affects business model content, structure, governance, and value logic (Aagaard & Tucci, 2024; Amit & Zott, 2001, 2021). Kulkow (2021), for example, shows how AI startups in healthcare create stakeholder value through improved access, reliability, and efficiency. Esau et al. (2025) explore the cognitive foundations of business model design and innovation, particularly in sustainable startups. While it does not focus explicitly on AI, it offers highly relevant insights for AI-driven entrepreneurial contexts, where uncertainty and paradoxes are common.

Finally, process studies could explore how AI shapes BMI ideation, iteration, and implementation (Aagaard & Tucci, 2024; Amit & Zott, 2021). For example, AI can act as a “co-pilot” in customer research or brainstorming, redefining how innovations are conceived and realized. Wang et al. (2025) directly connect digital technologies—including AI, machine learning, and big data analytics—to business model innovation. They offer a process model showing how new ventures design and evolve their business models to leverage digital technologies for societal and environmental impact.

Taken together, and illustrated in Table 1, the intersection of the three research areas (AI as opportunity, threat, and enabler of societal value) with BMI's antecedents, processes, and outcomes forms a 3×3 matrix of promising research opportunities.

4.2 | BMI and business/global unrest

Global unrest and business model innovation (BMI) are increasingly intertwined as socio-economic, political, and environmental disruptions create complex challenges for firms (Breier et al., 2021). Global unrest—manifested in

market volatility, political instability, inflation, supply chain breakdowns, and social tensions—disrupts traditional operations (Ivanov & Dolgui, 2022). Our review found few studies on this topic in leading management and entrepreneurship journals (see Supporting Information Appendices A2 and B2), yet existing research highlights BMI's vital role in enhancing organizational resilience and adaptability.

Bettiol et al. (2023) show how Italian SMEs adapted their models through servitization and customer-centric strategies to maintain stability amid shocks. Howell et al. (2018) demonstrate that BMI supports resilience to technological and institutional disruptions via frugal innovation and flexible models serving bottom-of-the-pyramid markets. Likewise, Siebold (2021) finds that social purpose organizations use BMI to sustain impact in turbulent contexts, while Yunus et al. (2010) illustrate how social and inclusive business models can address poverty and empower low-income communities.

Despite evidence of BMI's effectiveness in crisis response, limited research examines (a) how unrest can be leveraged as an opportunity for innovation and competitive advantage; (b) how BMI can mitigate threats from global unrest; and (c) how unrest can spur new models addressing grand challenges. Given recent disruptions—from the COVID-19 pandemic to the Russia-Ukraine conflict and rising geopolitical tensions—significant advances in this field are expected.

To guide future inquiry, we propose three research areas (a–c, i.e., global unrest as opportunity, threat and enabler of societal value) aligned with BMI's antecedents, processes, and outcomes (see Table 2).

Exploring antecedents involves understanding the contextual drivers of innovation—resources, capabilities, external trends, and stakeholder collaboration. Shukla and Sengupta (2021) emphasize stakeholder cooperation in initiating BMI during crises. Talebian et al. (2025) explain why and how global unrest triggers firms to pursue adaptive and collaborative business model innovations—balancing flexibility, stakeholder alignment, and resilience. Future studies should explore how firms strategically mobilize these antecedents to exploit instability.

Regarding outcomes, research should examine how crisis-driven adaptations, such as servitization (Bettiol et al., 2023), contribute to resilience, sustainability, and competitive advantage. Wang et al. (2025) demonstrate how digital technology-enabled BMI can provide stability, coordination, and impact under conditions of global unrest, turning volatility into opportunities for transformation. Further empirical work is needed to assess how socially and environmentally oriented BMI enhances both firm performance and societal stability.

Finally, the processes of BMI under instability deserve closer study. Although scholars note the importance of digital transformation and platform innovations (Fu et al., 2023), little is known about how firms ideate, prototype, and implement new models. Esau et al. (2025) explain how entrepreneurial cognition enables firms to design resilient business models that integrate purpose and adaptability under volatile, high-stress conditions. Future research could investigate how businesses integrate stakeholder input, iterate designs, and manage partnerships to embed innovation in their operations. Understanding these processes can clarify how organizations build adaptive capacity to manage risks and seize opportunities during turbulent times.

4.3 | BMI and climate change

Climate change refers to “a change of climate attributed directly or indirectly to human activity that alters the global atmosphere” (IPCC, 2018a). Human activities such as land use, transport, tourism, and production systems drive these changes, and rising populations and consumption will exacerbate them without major behavioral and policy interventions (Gibson et al., 2024). Global climate policy aims to limit warming to below 1.5°C above pre-industrial levels (IPCC, 2018b), yet record temperatures raise doubts about achieving this goal (Copernicus, 2023). Exceeding 1.5°C will make mitigation harder and costlier; at 2°C, the world faces severe cascading effects on ecosystems, resources, food systems, and cities (IPCC, 2018b). For the EU and UK, adaptation costs are estimated at €40 billion annually for a 1.5°C scenario, rising to €80–120 billion at 2°C—excluding human and social losses (European Environment Agency, 2024).



Business model innovation (BMI) has been recognized as a lever for addressing environmental and social challenges (Boons & Lüdeke-Freund, 2013). However, only 23 papers we identified explicitly link BMI and climate change (see Supporting Information Appendices A3 and B3). To advance this field, we propose a 3×3 framework for research at the intersection of climate change and BMI (see Table 3), paralleling those for AI and global unrest.

In examining antecedents of BMI, prior studies highlight barriers and drivers of sustainable innovation across industries (Coffay et al., 2024; Hansson et al., 2023). While environmental and regulatory pressures motivate some firms, internal business cases—such as cost savings or new revenue opportunities—often determine adoption (Schaltegger et al., 2012). Talebian et al. (2025) conceptualize how firms use BMI as a robustness mechanism to navigate climate-related volatility and environmental uncertainty. Future research should explore how firms align internal functions and incentives to embed climate-focused BMI within core innovation processes.

Regarding outcomes, most climate-related BMI studies remain conceptual and seldom empirically assess economic, environmental, or societal performance (Snihur & Bocken, 2022). Rare empirical work shows that models like rentals and pay-per-use can significantly reduce environmental impact compared to product sales (Sarasini et al., 2024; Tukker, 2004). Yet, these benefits depend on intentional design that anticipates impacts on logistics and consumer behavior. Wang et al. (2025) show how digital and AI-enabled business models can accelerate climate change mitigation and adaptation by expanding firms' capacity to generate systemic, cross-sectoral impact. Future studies could develop better indicators and assessment tools to measure BMI's real climatic effects across value chains.

In studying processes, scholars have applied the dynamic capabilities lens to examine how established firms build and manage sustainable BMI (Bocken & Konietzko, 2022). Research identifies institutional, strategic, and operational barriers and enablers (Bocken & Geradts, 2020). Emerging ventures, particularly in the sharing economy, can act as agents of institutional change (Zvolnska et al., 2019). Esau et al. (2025) explain how cognitive diversity and paradoxical thinking enable entrepreneurs to design viable climate- and sustainability-focused business models under conditions of uncertainty. Future work should identify the virtuous and vicious cycles within BMI (Snihur & Bocken, 2022) and explore how businesses can overcome structural barriers to accelerate climate-positive transformation.

5 | DISCUSSION AND FUTURE RESEARCH OPPORTUNITIES

In this article, we present a novel framework to advance research on business model innovation (BMI) in the context of grand challenges. Our $3 \times 3 \times 3$ framework integrates three grand challenges (ABC; see Tables 1–3); three key aspects of business models (antecedents, outcomes, and processes; see rows in Tables 1–3); and three possible intersections between BMI and grand challenges (GC as an enabler of BMI opportunities, BMI to mitigate threats from BC, and BMI as an enabler of societal value; see columns in Tables 1–3). The “BMI-GC” concept is introduced to guide future research and practice in the area of BMI as a driver for, and response to addressing grand challenges.

We emphasize research at the intersection of *BMI and AI* and offer complementary perspectives on how firms design, adapt, and cognitively manage innovation under technological and societal pressures. Wang et al. (2025) develop a process model showing how ventures leverage AI and other digital technologies through stages of accessing, embedding, strengthening, and expanding business models to scale their impact across ecosystems addressing grand challenges. Talebian et al. (2025) link environmental dynamism—often amplified by AI-driven technological change—to firms' engagement in robust action and corresponding types of BMI, illustrating how uncertainty and turbulence shape the novelty and scope of business model transformation. Esau et al. (2025) adopt a cognitive lens to explain how founders' mindsets and value orientations influence sustainable BMI trajectories, emphasizing the importance of paradoxical thinking for balancing commercial and societal objectives. Collectively, these studies advance understanding of how AI-enabled technologies, dynamic environments, and entrepreneurial cognition interact to drive business model innovation for competitive advantage and societal impact.

We offer valuable insights into how *BMI enables organizations to navigate business and global unrest*. Talebian et al. (2025) conceptualize how firms operating under dynamic and turbulent environments engage in robust, multi-stakeholder action that fosters adaptive forms of BMI. Wang et al. (2025) demonstrate how digital technologies, including AI and data analytics, support business model design that enhances coordination, resilience, and social impact across ecosystems facing systemic disruption. Esau et al. (2025) complement these perspectives by showing how founders' cognitive frames—particularly paradoxical thinking—enable ventures to develop viable and sustainable business models amid uncertainty and constraint. Together, these studies deepen the understanding of how organizations combine technological, strategic, and cognitive adaptation to innovate business models capable of thriving amid global instability.

Last but not least, we connect *BMI to organizational responses to climate change* across technological, strategic, and cognitive dimensions. Wang et al. (2025) show how digital and AI-enabled technologies facilitate business model design and scaling to address environmental challenges, outlining a process through which firms integrate, strengthen, and expand sustainability-oriented solutions across ecosystems. Talebian et al. (2025) conceptualize how firms navigate climate-induced environmental dynamism through robust action—collaborative, experimental, and adaptive strategies that enable the development of resilient business models under uncertainty. Esau et al. (2025) complement these insights by revealing how entrepreneurs' cognitive frames shape sustainable BMI trajectories; ventures embracing paradoxical thinking are more likely to balance ecological, social, and economic goals effectively. Together, these studies advance understanding of how firms leverage digital technologies, strategic adaptability, and cognitive diversity to design and implement business models that both mitigate and adapt to climate change, contributing to long-term sustainability transitions.

Using our ABC-framework, we show that the three articles in this Special Section address important questions, but also leave plenty of room for future discovery. Specifically, we identify over 27 research questions that offer fertile ground for rigorous scholarly inquiry, with the *Strategic Entrepreneurship Journal* positioned as a key venue and central platform for such work. The Special Section features three papers that exemplify innovative approaches to this nexus, showcasing the promise of future studies while revealing opportunities to deepen understanding of the strategic role of entrepreneurial BMI in tackling critical global issues.

Plenty of new insights await to be discovered, and many fascinating cases of strategic entrepreneurship deployed toward using BMI to solve grand challenges await to be examined, as shown by new startups and corporate initiatives all over the world. In the Netherlands, the company MUD Jeans, for example, started with a leasing concept as a circular model to retrieve jeans and reclaim materials after use (see Bocken et al., 2018). The business model currently co-exists with a product sales and take-back model. MUD Jeans contain up to 40% post-consumer recycled cotton with the goal to go to 100% (MUD Jeans, 2024). In Japan, Kuradashi sells food that is still edible, which otherwise would be thrown away, using an app and platform for consumers. Food is delivered to consumers' homes and a social aspect of the business model is that donations are made to good causes, such as food banks, in Japan (Circular X, 2024). In India, Inditech in collaboration with the Indian Association of Pediatrics and the South Asia Pediatric Association is developing and deploying an AI-driven business model to provide education (webinars, tutorials, online lectures, etc.) and tools to pediatricians and students of medicine at scale (Kowalewski et al., 2025). The venture aims to provide various novel services, such as “university as a service,” early-stage detection of critical illnesses in children through a red-flag alert system, and expert advice—in countries with massive populations, such as India, Pakistan, or Bangladesh, where expert pediatricians are in scarce supply, leading to serious bottlenecks and under-treatment of patients. The case exemplifies the importance of combining a partnership-driven approach (embracing government institutions, corporations, and tech companies), cutting-edge technologies, and smart business model design to address grand challenges effectively at scale.

Strategic entrepreneurship, when applied through BMI, reframes global challenges as opportunities. By doing so, entrepreneurial individuals and organizations pursue projects with the strategic goal of serving not only their shareholders but also society as a whole. In other words, the goal of their BMI efforts is to enhance the total value created for all—beyond immediate business model stakeholders such as the focal firm, the firms' customers, suppliers, and



partners. This is taking the notion of total value creation (Brandenburger & Stuart Jr, 1996) to a new level of meaning. BMI applied to grand challenges represents an important way of thinking and acting both entrepreneurially as well as strategically, emphasizing collaborative, systemic approaches to value creation.

5.1 | Expanding the boundaries of business model innovation

Linking BMI to grand challenges thus pushes the theoretical boundaries of the business model construct. Zott et al. (2011) identified four key themes in business model research: boundary-spanning analysis, a system-level approach, activity focus, and the creation and capture of value. Examining BMI in the context of grand challenges further expands these themes by: (1) *boundary-spanning*: recognizing humanity and nature as key stakeholders (see also Stubbs & Cocklin, 2008); (2) *system-level approach*: positioning business models within complex business ecosystems that are embedded in ecological, political, and societal contexts; (3) *activity focus*: evaluating the broader socio-economic impact and ecological consequences of business model activities; and (4) *total value creation*: advancing beyond the traditional game-theoretic framework to emphasize holistic, multi-stakeholder value creation.

The democratization of innovation underscores the potential of BMI in tackling grand challenges. Unlike product and technological innovations that rely heavily on elite scientists and research institutions, BMI can be pursued by virtually anyone. The discovery-driven and often methodical and experimental nature of BMI allows individuals and firms to test, iterate, and scale solutions in a lean and flexible manner (Chesbrough & Tucci, 2020; Zott & Amit, 2024), empowering a broader array of contributors to engage in solving pressing societal issues.

5.2 | Future research agenda

To advance the understanding and application of BMI in addressing grand challenges, a future strategic entrepreneurship research agenda may want to tackle and explore the following overarching themes that cut across the domains described above.

Measurement and assessment: Research is needed to develop and validate robust metrics that assess the impact of BMI on addressing grand challenges, including environmental, economic, and social outcomes. The development of metrics to assess BMI's effectiveness in addressing grand challenges remains an important area for research.

Digital transformation: As digital technologies intersect with each of the grand challenges, understanding the role of digital tools (e.g., LLMs, blockchain, the Internet of Things, advanced data analytics) in enabling BMI across the ABC classification is crucial. Exploring how BMI intersects with digital transformation, supply chain resilience, and sustainable practices will provide valuable insights for managers, policymakers, and scholars aiming to harness the potential of BMI to address some of the most pressing challenges of our time.

Collaborative ecosystems: Investigating the role of partnerships and multi-stakeholder collaboration in enhancing BMI efficacy. Research can focus on how collaborative business ecosystems contribute to BMI, especially when addressing complex challenges that require coordinated actions across industries. Future research should address the interplay between technological advances, regulatory landscapes, and stakeholder interests within each domain.

Some cross-domain sample research questions include, but are not limited to, the following: How can the success of BMI be measured with respect to its impact on grand challenges, such as AI ethics, economic resilience, and environmental sustainability? What role does digital transformation play in facilitating BMI to address AI, global unrest, and climate change? How can collaborative ecosystems involving multiple stakeholders enhance the efficacy of business models aimed at tackling grand challenges?

6 | CONCLUSION

By embracing these research pathways, scholars can contribute to a deeper theoretical and practical understanding of BMI spurred by grand challenges, and as a means to address them. This Special Section aims to inspire and initiate significant new work in this area, with the ambition to demarcate strategically important directions for future research within the Strategic Management Society's six big questions in strategic management,² especially questions 1 ("How does the increased focus on sustainability [environmental, social, and economic] impact business models of the future?"); 2 ("How does disruptive technology and the explosion of access to data impact firm strategy, human capital management, and society?"); and 4 ("What impact do globalization pressures have on firm strategy, entrepreneurial opportunities, interorganizational arrangements, and shifts in the prominence of different economies?"). Clearly, regarding these big questions an increased focus on sustainability (environmental, social, and economic) necessitates that future business models integrate ESG considerations, cooperative strategies, and corporate social responsibility, aligning profitability with societal impact, addressing B and C in our ABC classification. Disruptive technology and the explosion of data access reshape firm strategy by driving digital transformation, necessitating strategic agility in human capital management and fostering greater societal connectivity, addressing A and B. Additionally, deglobalization pressures compel firms to rethink global strategies, fostering entrepreneurial opportunities through localized and decentralized business models while navigating complex interorganizational arrangements and regional economic shifts, addressing the B in our ABC classification.

The BMI-GC concept is proposed as an important driver for future research and practice. We highlight the substantial research potential within BMI as a means of addressing grand challenges, offering scholars a robust conceptual toolkit for advancing both theoretical and practical understanding in the fields of strategic management and entrepreneurship.

ACKNOWLEDGMENTS

We would like to thank Melissa Graebner, former *SEJ* co-editor and liaison editor for this Special Section, for her unwavering support. We also thank all current and past co-editors of the *SEJ* who promoted and supported the creation of this Special Section, notably (in alphabetical order) Gary Dushnitsky, Peter Klein, Yong Li, and David Sirmon. They provided us with excellent feedback. All remaining errors in this manuscript are ours.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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ENDNOTES

¹ Berkeley EECS website. Q&A: The future of artificial intelligence. <https://people.eecs.berkeley.edu/~russell/research/future/q-and-a.html>.

² See <https://www.strategicmanagement.net/strategic-insights/>.



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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Bocken, N. M. P., Heidenreich, S., Spieth, P., Tucci, C., & Zott, C. (2025). Innovating for impact: Harnessing business model innovation to tackle grand challenges. *Strategic Entrepreneurship Journal*, 19(4), 665–681. <https://doi.org/10.1002/sej.70010>